

CS & IT ENGINEERING



Digital Logic

Sequential Circuits

Lecture No 02



By- Aditya sir

Recap of Previous Lecture



Topic



Intro to Sequential Circuits

→ Latches

→ S-R Latch. (NAND gate)

→ Problems

Topics to be Covered



Topic



Combinational Circuits

- Parallel Adder (Ripple Carry Adder)
- Look Ahead Carry Adder
- Multipliers



About Aditya Jain sir



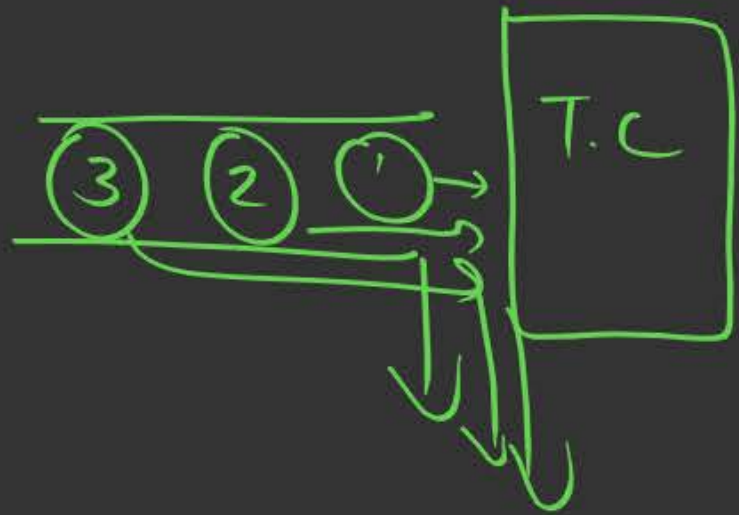
1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



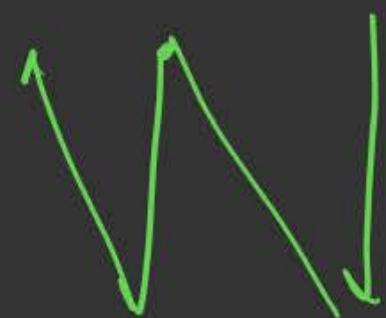


Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

→ Serial Adder



One by one (in Sequence)



Escool Days

$$\begin{array}{r} 011 \rightarrow \text{Carry} \\ 7275 \\ + 4397 \\ \hline \end{array}$$

$$\underline{11672} \rightarrow \text{Sum}$$

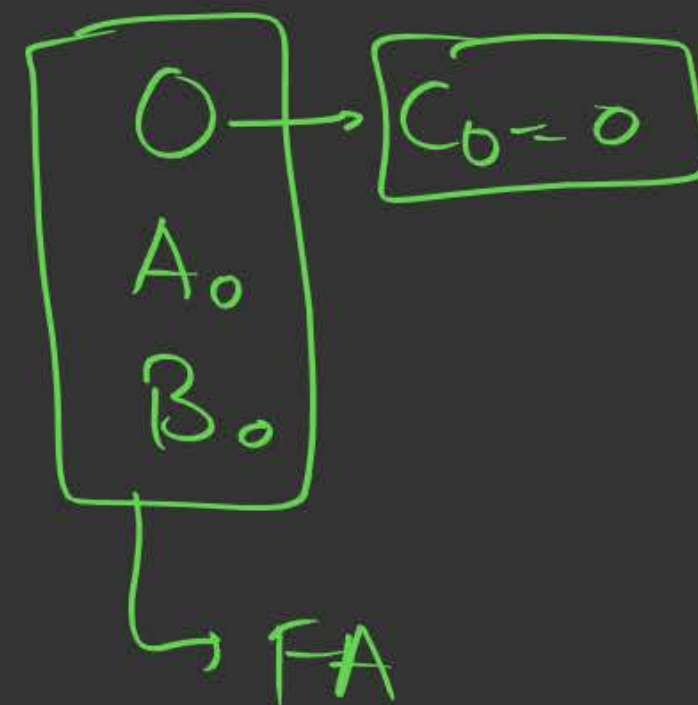
$n=4$



$$\begin{array}{r} C_3 \quad C_2 \quad C_1 \\ A_3 \quad A_2 \quad A_1 \quad A_0 \\ + B_3 \quad B_2 \quad B_1 \quad B_0 \end{array}$$

HA

$$\underline{C_4 \quad S_3 \quad S_2 \quad S_1 \quad S_0}$$



In genal,

$$n \text{ bits} \Rightarrow \text{FA} \rightarrow (n-1)$$

$$\text{HA} \rightarrow 1$$

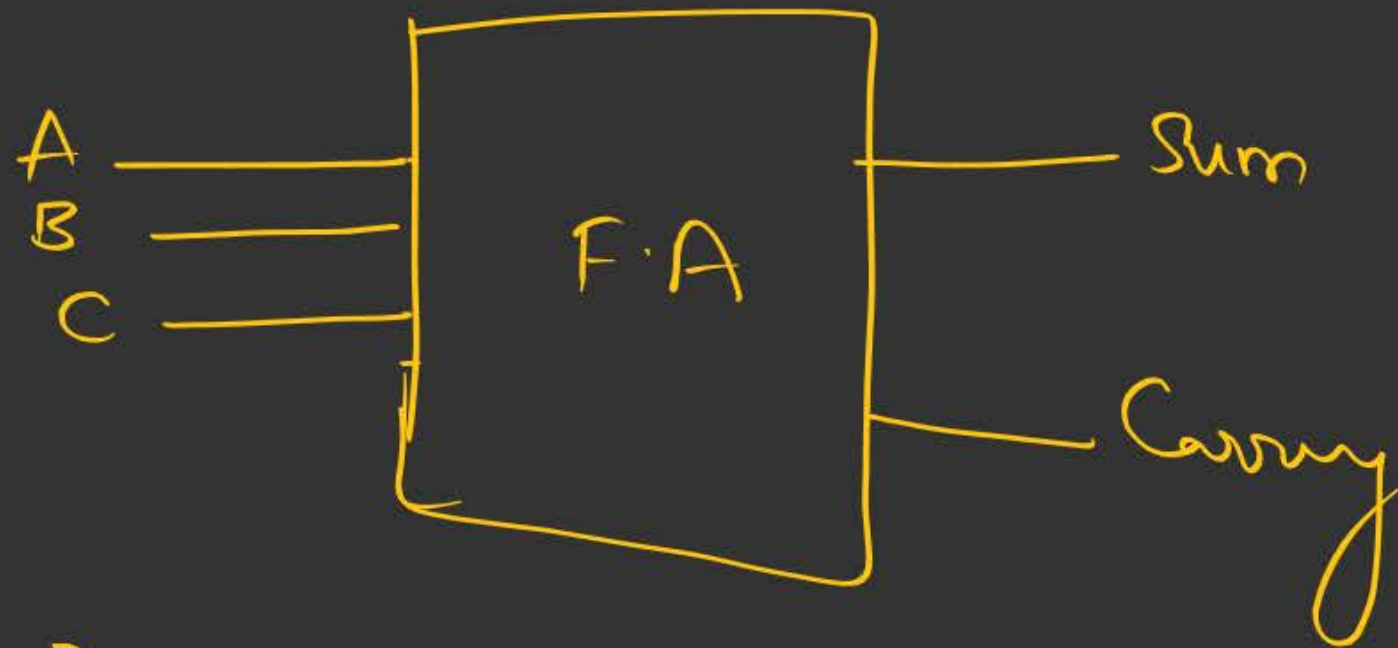
$$\underline{n=4}, \text{FA}=3$$

$$, \text{HA}=1$$

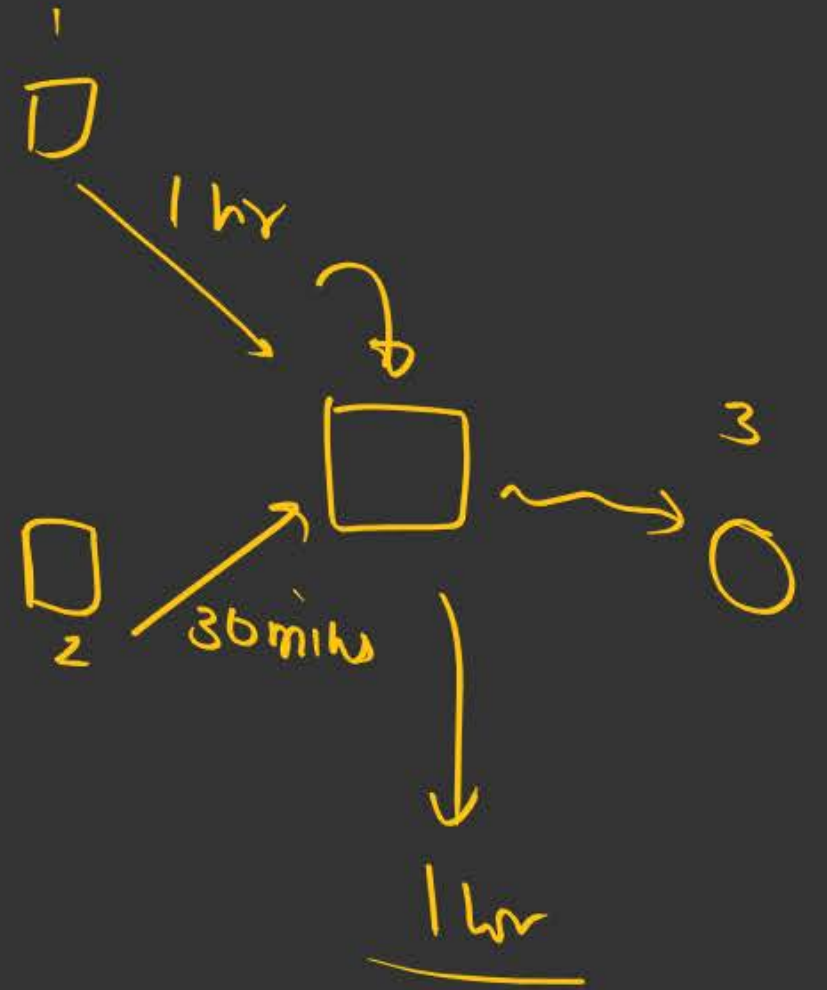
$$\underline{n=5}, \text{FA}=4$$

$$, \text{HA}=1$$

$d_s + d_c \times$



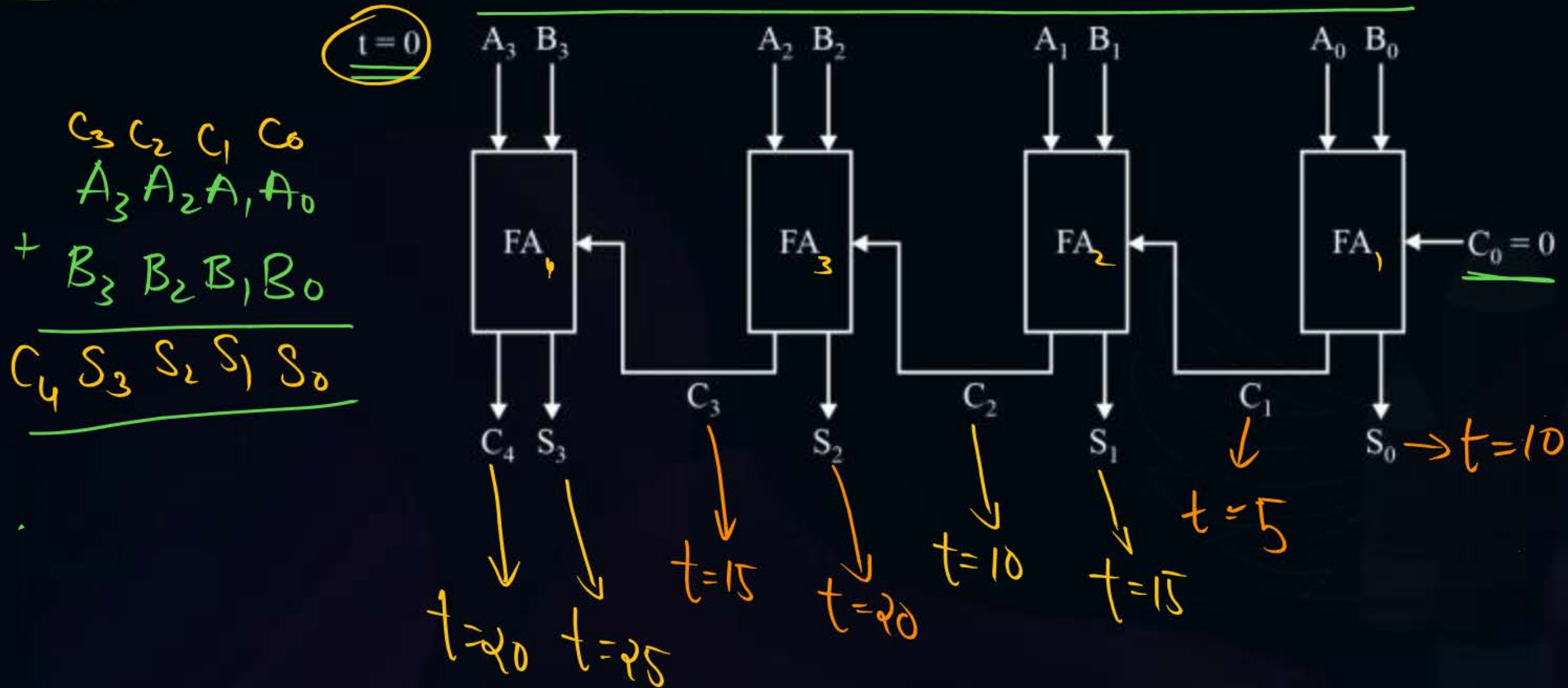
$$\text{Delay}_{FA} = \max\{d_{\text{sum}}, d_{\text{carry}}\}$$





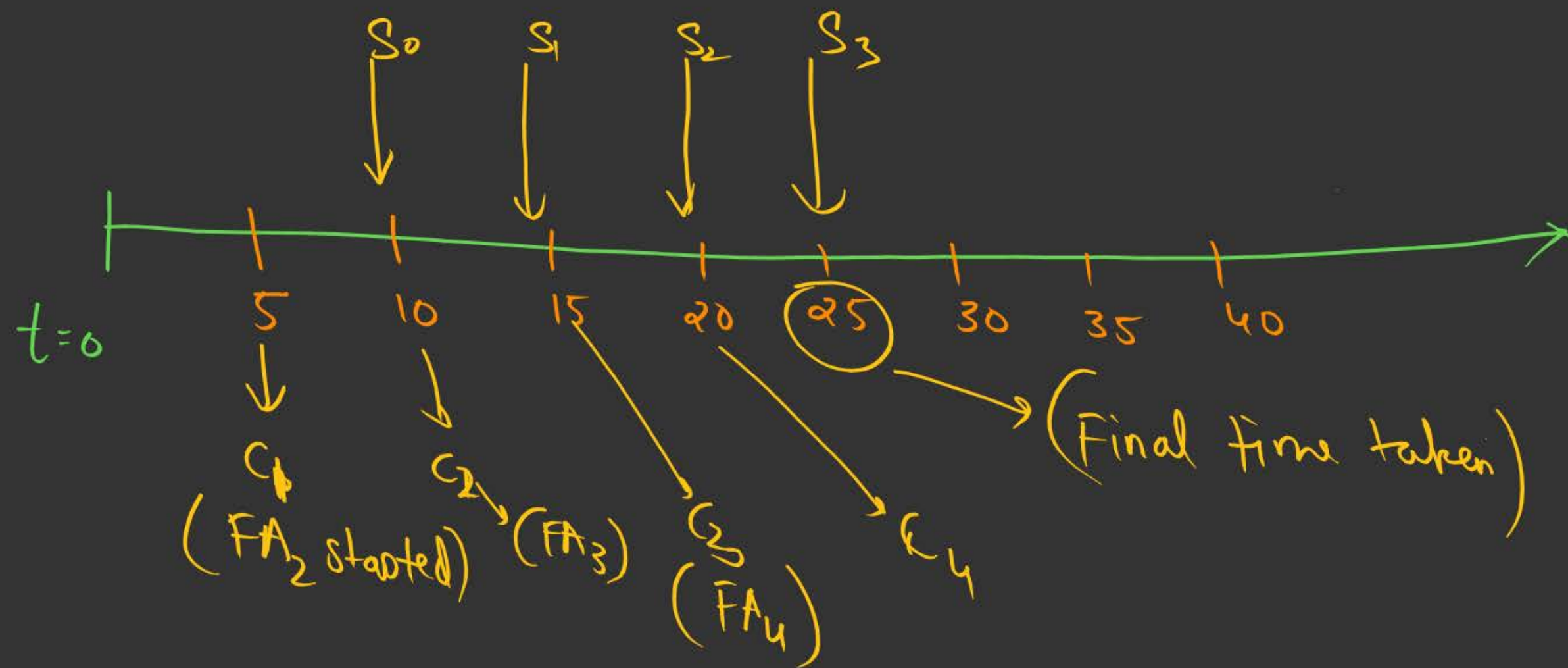
Topic : PARALLEL ADDER, LACA

PARALLEL ADDER (Ripple Carry Adder):



Suppose
 $T_g = 10 \text{ sec}$
 $T_c = 5 \text{ sec}$

Timeline (for per eg)





Topic : PARALLEL ADDER, LACA

n-bit Parallel Adder:

- ✓ 1. $(n-1)$ FA + 1 HA
- ✓ 2. n - FA
- ✓ 3. $(2n-1)$ HA + $(n-1)$ OR

Delay:

$$T = \underbrace{(n-1) T_{\text{carry}}}_{\text{Carry propagation delay}} + \underbrace{\text{Max} \{T_{\text{sum}}, T_{\text{Carry}}\}}_{\text{Sum delay}}$$

$$1 \text{ FA} = \underbrace{2 \text{ HA} + 1 \text{ OR}}$$

$$PA = (n-1) \text{ FA} + 1 \text{ HA}$$

$$= (n-1) [2 \text{ HA} + \text{OR}] + 1 \text{ HA}$$

$$= \underline{2(n-1) \text{ HA}} + (n-1) \text{ OR} + \underline{1 \text{ HA}}$$

$$= \underline{(2n-1) \text{ HA}} + (n-1) \text{ OR}$$

(V. Imp)

eg:- $T = \underbrace{(n-1)}_{\text{wavy line}} T_c + \max(T_s, T_c)$

Check:

$$T_c = 5$$

$$T_s = 10$$

$$n = 4$$

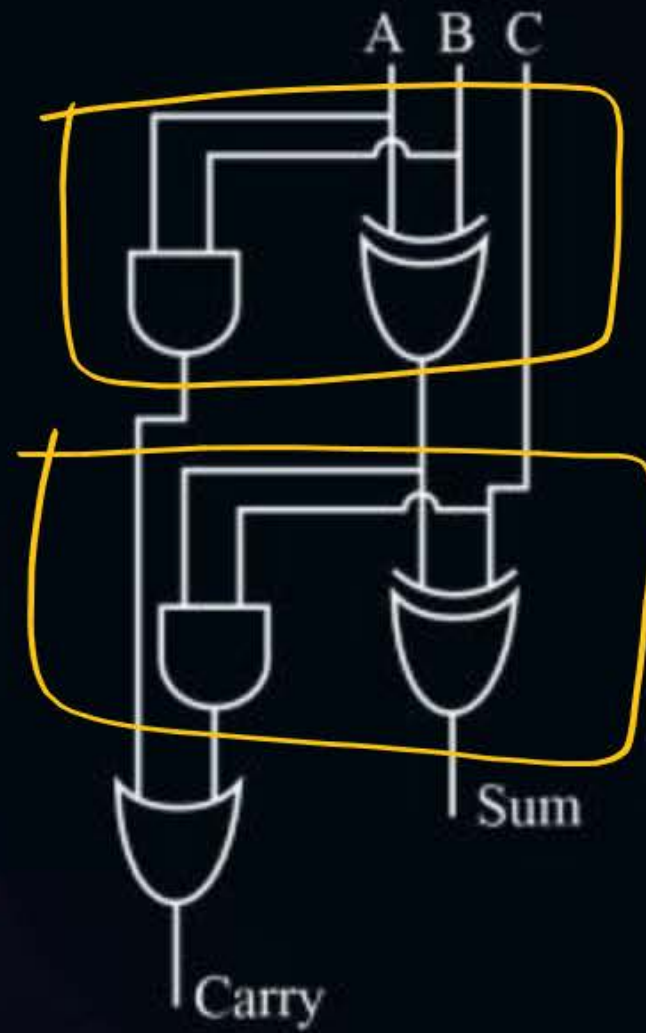
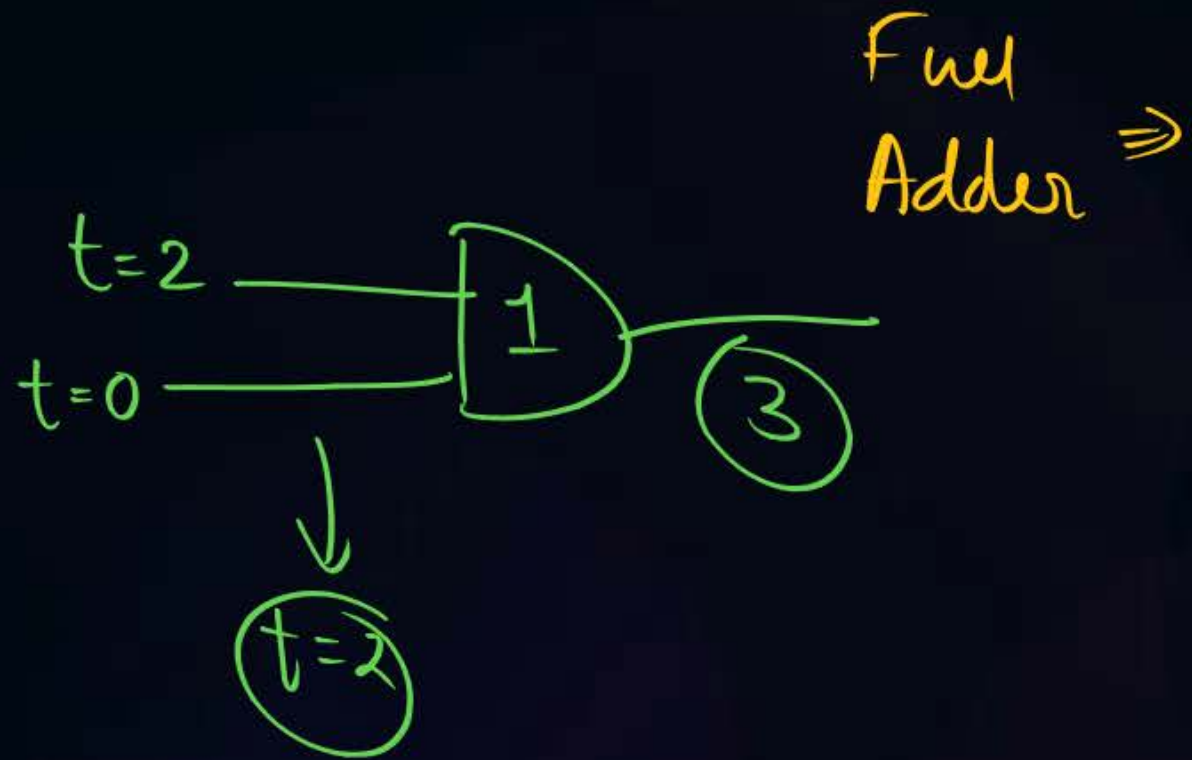
$$= (4-1) \times 5 + \max(10, 5)$$

$$= 3 \times 5 + 10$$

$$= \boxed{25} \text{ } ^{\circ}10 \text{ } \smile$$



Topic : PARALLEL ADDER, LACA



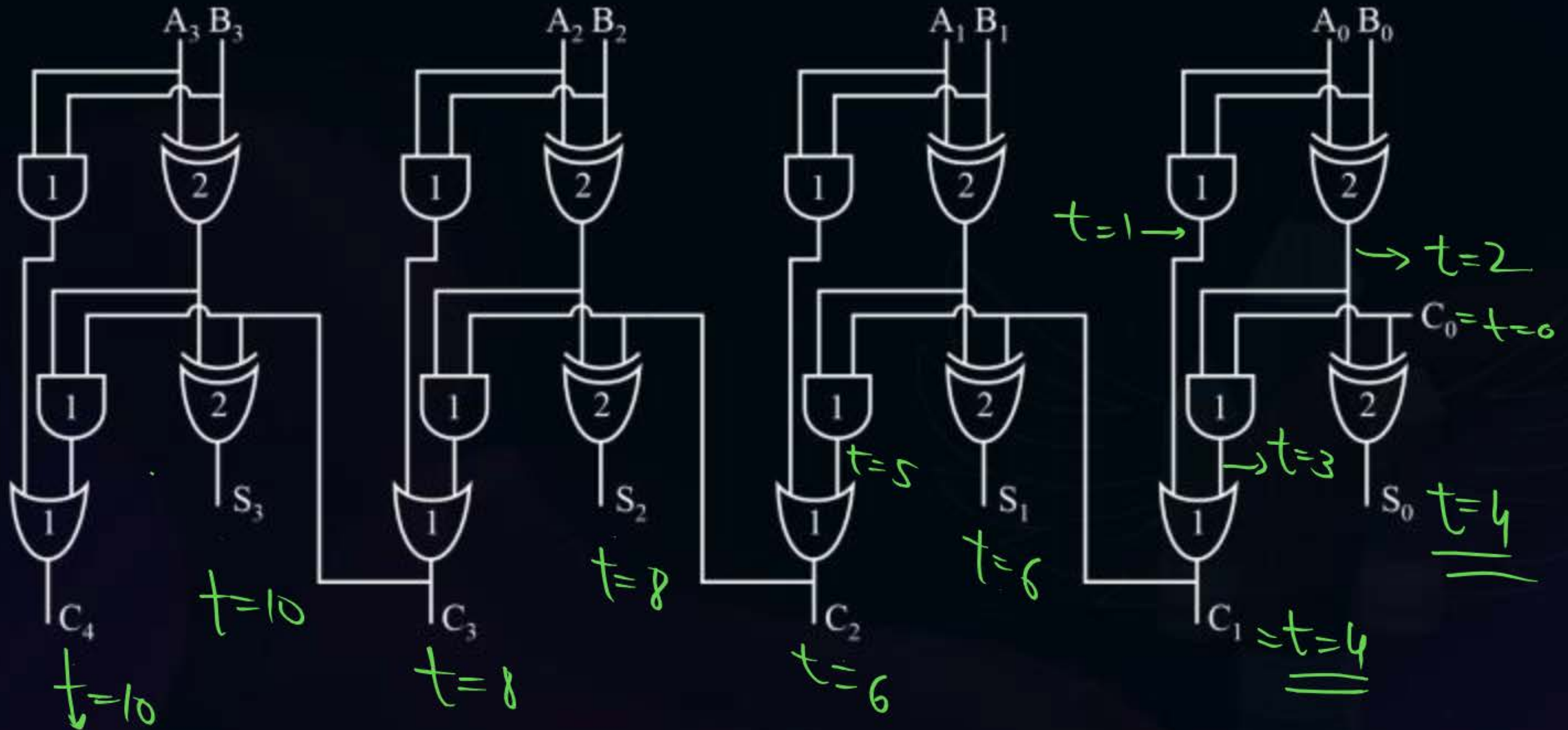


Topic : PARALLEL ADDER, LACA

PARALLEL ADDER [4-bit]:

given: $T_{XOR} = 2$
 $T_{OR} = 1$, $T_{AND} = 1$

$t = 0$





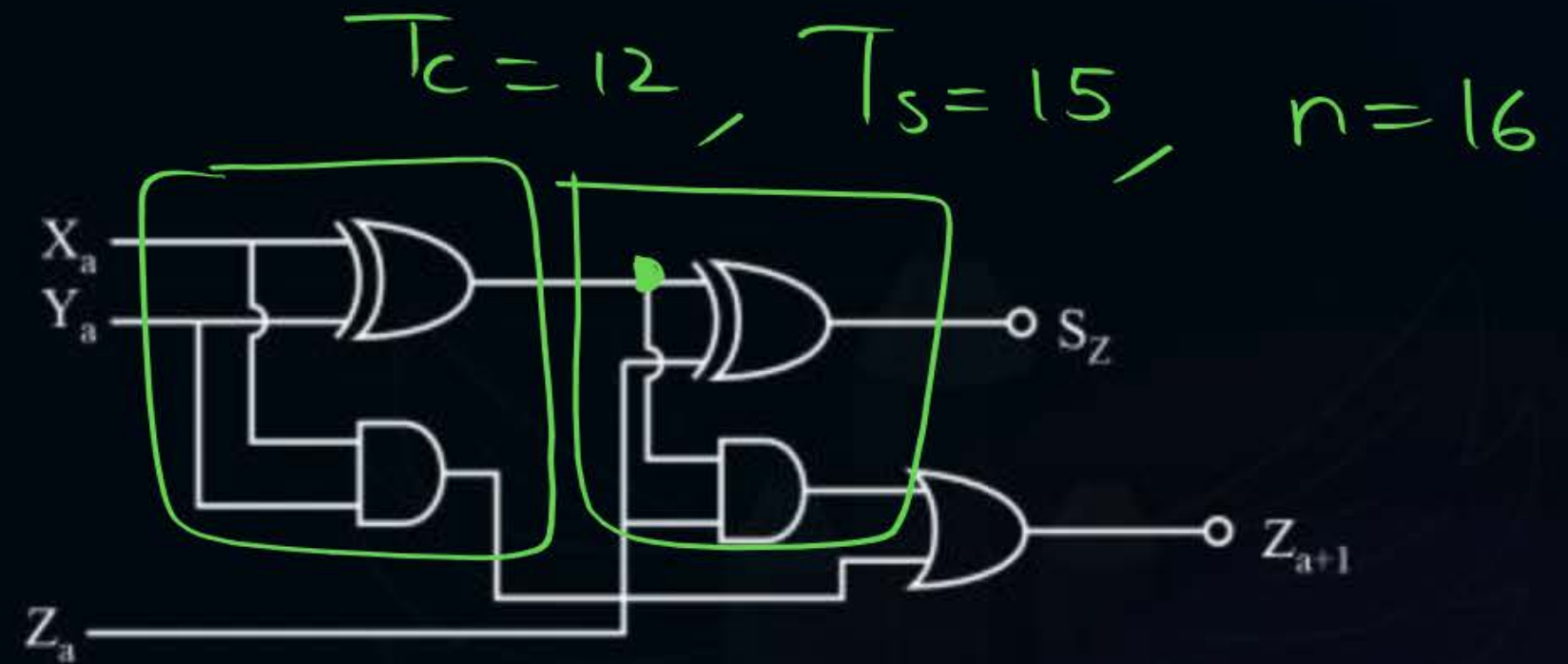
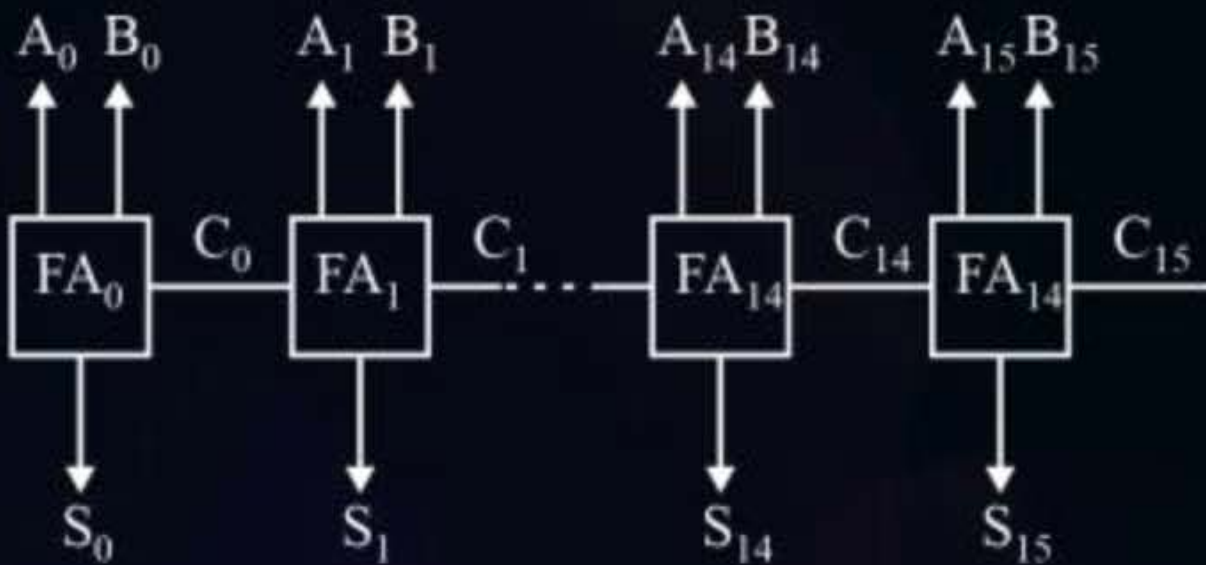
Topic : PARALLEL ADDER, LACA

$$T = (n - 1) \{T_{\text{AND}} + T_{\text{OR}}\} + \text{Max} \{T_{\text{Sum}}, T_{\text{Carry}}\}$$

$$T = (4 - 1) \{1 + 1\} + \text{Max} \{4, 4\}$$

$$T = 3 \times 2 + 4 = 10 \text{ NS}$$

#Q. A 16-bit ripple carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry-propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be _____.



70%
↓
75%



Topic : PARALLEL ADDER, LACA

Solution:

$$T = (n - 1) T_{\text{Carry}} + \text{Max} \{T_{\text{Sum}}, T_{\text{Carry}}\}$$

$$T = (16 - 1) \times 12 \text{ ns} + \text{Max} \{15 \text{ ns}, 12 \text{ ns}\}$$

$$T = 15 \times 12 + 15 = \underline{\underline{195 \text{ ns}}} \quad \checkmark$$



#Q. A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit ripple-carry binary adder is implemented by using full adders. The total propagation time of this 4-bit binary adder in microseconds is,

int()

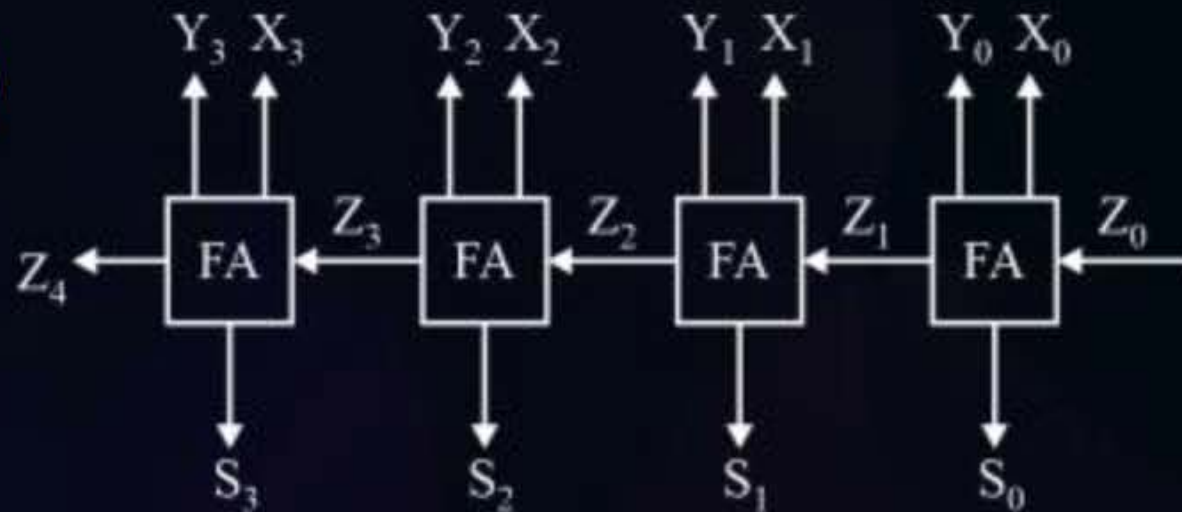


Figure-I

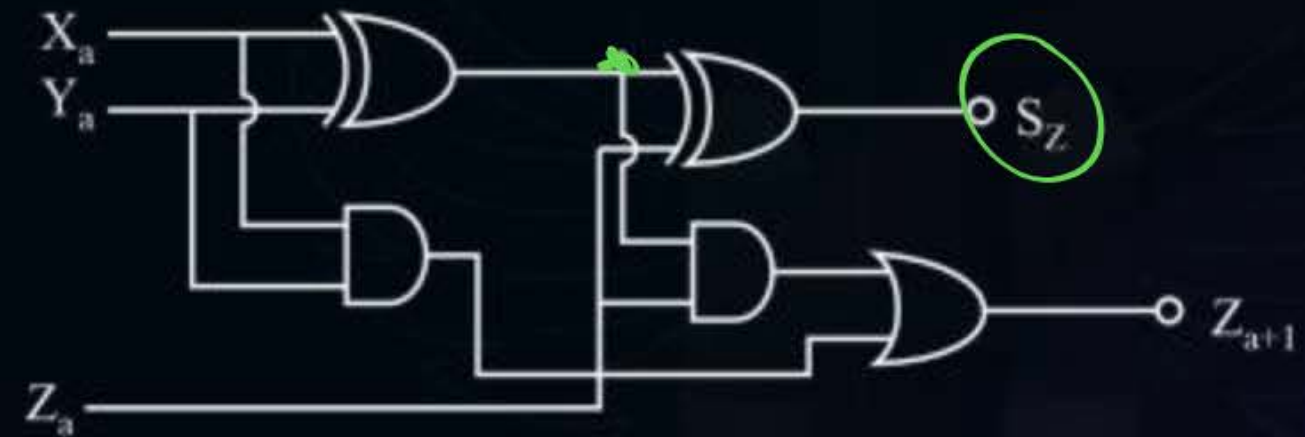


Figure-II



Topic : PARALLEL ADDER, LACA

Solution:

given $T_{XOR} = 2.4 \mu s$

$T_{AND} = T_{OR} = 1.2 \mu s$

$n = 4$

$T_{XOR} = 2 \times T_{AND}$
 $= 2 \times 1.2$
 $= 2.4$

$$T_{Sum} = 2 T_{XOR}$$

$$T_{Carry} = T_{XOR} + T_{AND} + T_{OR}$$

$$2.4 + 1.2 + 1.2$$

$$T = (n-1) [T_{OR} + T_{AND}] + \max(T_s, T_c)$$

$$T = 3 \times \{1.2 + 1.2\} + 2 \times 2.4$$

$$T = 3 \times 2.4 + 4.8$$

$$T = 12 \mu s$$

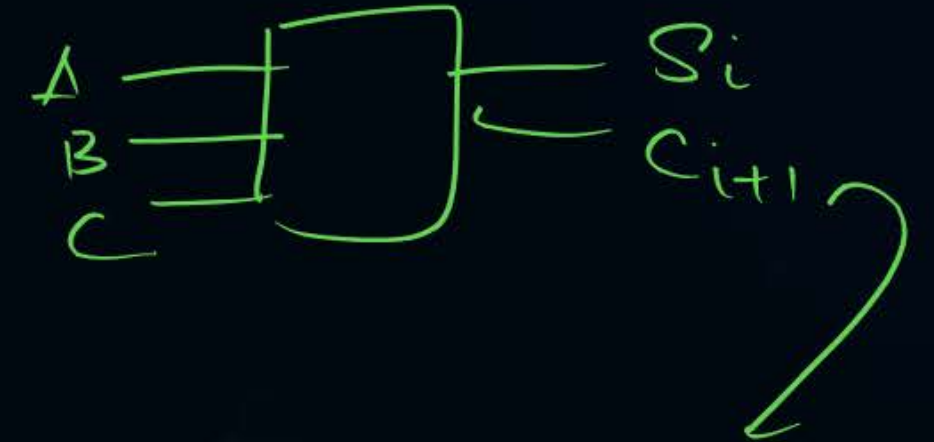
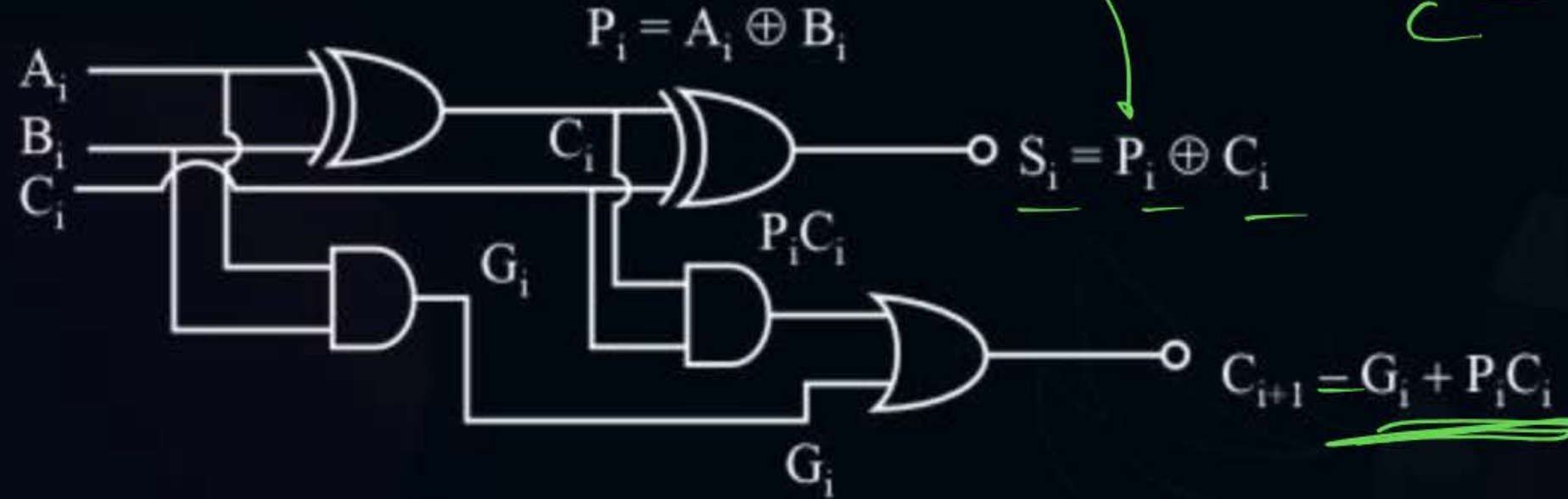
$$\downarrow \downarrow$$
$$\max(2 \times 2.4, 4.8)$$



Topic : PARALLEL ADDER, LACA

Look Ahead Carry Adder (LACA):

Fastest Adder among all the adder.



$$\underline{C_{i+1} = G_i + P_i \cdot C_i}$$

$P_i \rightarrow$ Carry propagating term

$G_i \rightarrow$ Carry generating term



Topic : PARALLEL ADDER, LACA

3 bit

$$P_i = A_i \oplus B_i$$

$$P_0 = A_0 \oplus B_0$$

$$P_1 = A_1 \oplus B_1$$

$$P_2 = A_2 \oplus B_2$$



$$S_i = P_i \oplus C_i$$

$$S_0 = P_0 \oplus C_0$$

$$S_1 = P_1 \oplus C_1$$

$$S_2 = P_2 \oplus C_2$$



$$G_i = A_i B_i$$

$$G_0 = A_0 B_0$$

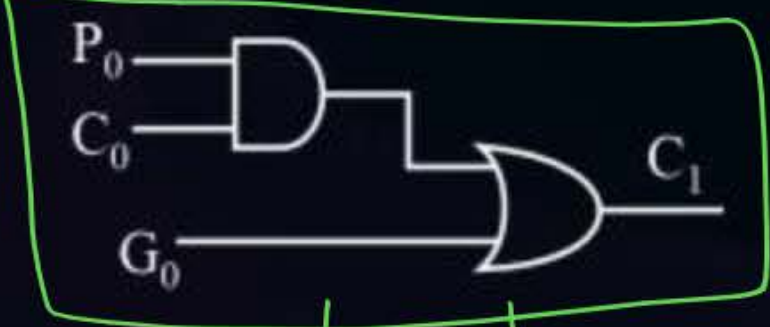
$$G_1 = A_1 B_1$$

$$G_2 = A_2 B_2$$



$$C_{i+1} = G_i + P_i C_i$$

$$C_1 = G_0 + P_0 C_0$$

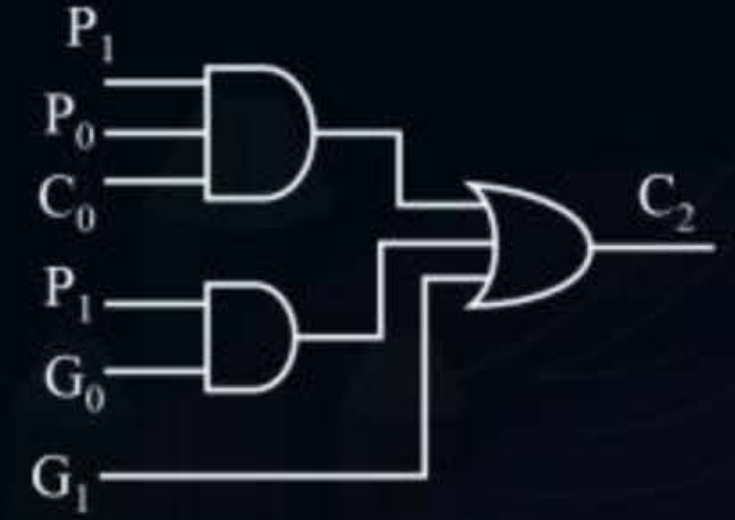


$$\begin{array}{r} A_2 A_1 A_0 \\ + B_2 B_1 B_0 \end{array}$$

$$C_2 = G_1 + P_1 C_1$$

$$C_2 = G_1 + P_1 [G_0 + P_0 C_0]$$

$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$



$$C_3 = G_2 + P_2 C_2$$



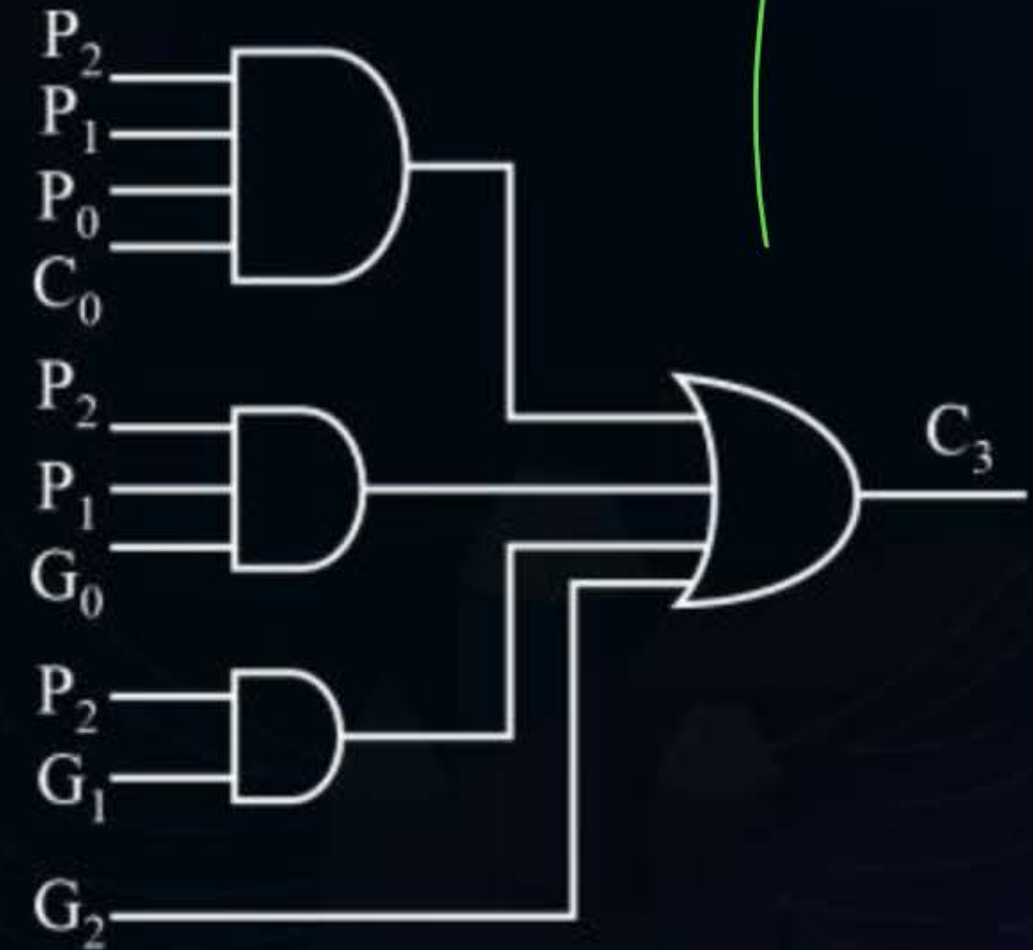
Topic : PARALLEL ADDER, LACA

$$C_3 = G_2 + P_2 C_2$$

$$C_3 = G_2 + P_2 [G_1 + P_1 G_0 + P_1 P_0 C_0]$$

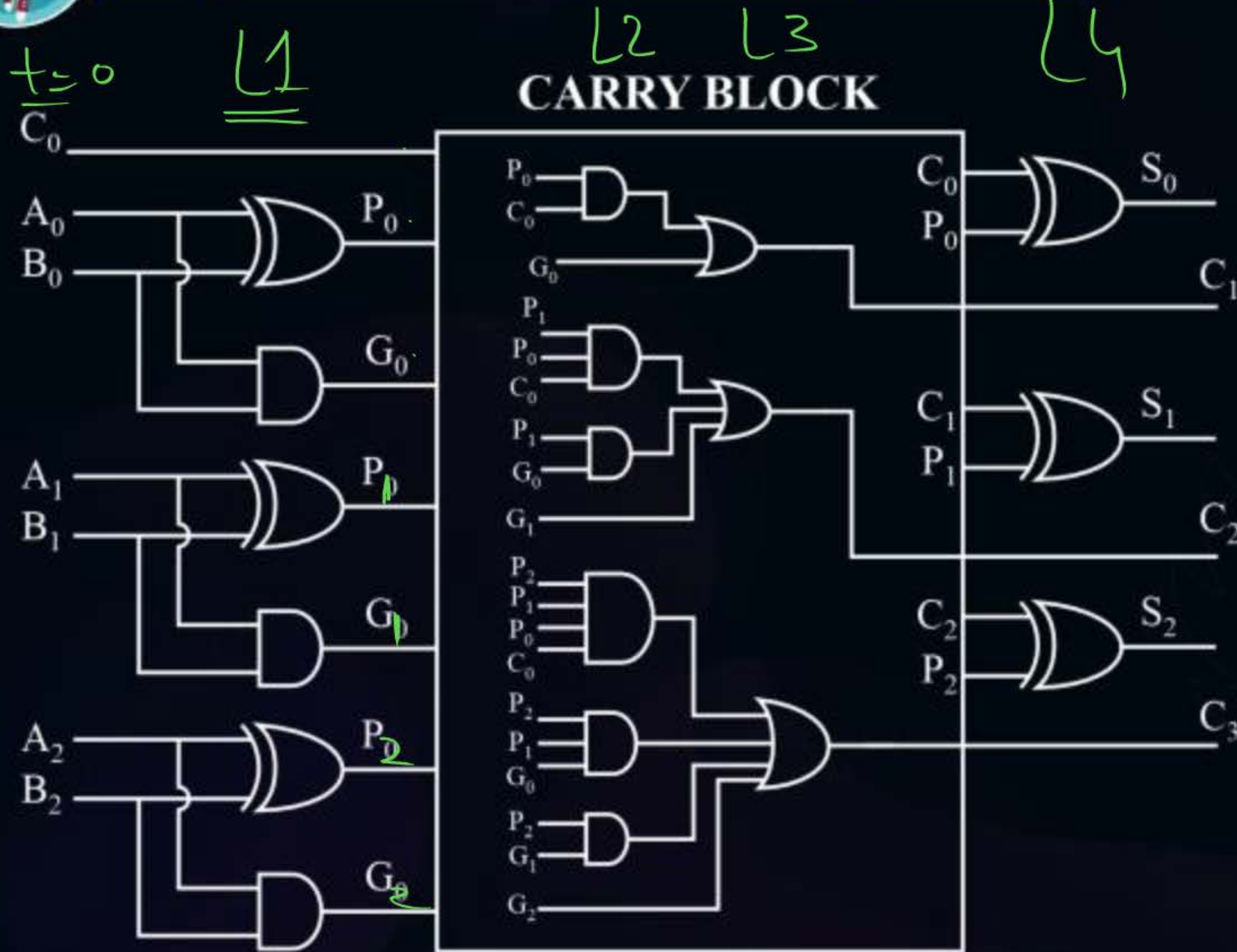
$$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

[AND - OR]
L1 L2





Topic : PARALLEL ADDER, LACA



Carry Block will be always two level implementation.

Delay of level = T

Delay for Carry Block: $2T$

For entire circuit: $4T$

$C_1 \rightarrow \text{AND} = 1, \text{OR} = 1$

$C_2 \rightarrow \text{AND} = 2, \text{OR} = 1$

$C_3 \rightarrow \text{AND} = 3, \text{OR} = 1$



Topic : PARALLEL ADDER, LACA

LACA
'n' bit ~~LADA~~:

For Carry Block:

$$\text{No. of AND GATE} = \frac{n(n+1)}{2}$$

$$\text{No. of OR GATE} = n$$

$$(1+2+3 \dots n)$$

$n \Rightarrow n$ AND
1 OR

$C_i \rightarrow i$ AND gates
1 OR gate

n-bit

$\Rightarrow C_1 \quad C_2 \quad C_3$

Total AND gates = $1 + 2 + 3 = 6$
in LACA.

" OR = $1 + 1 + 1 = 3$



Topic : PARALLEL ADDER, LACA

Multiplier:

2-bit:

$$A \rightarrow A_1 A_0$$

$$B \rightarrow B_1 B_0$$

$A \times B$:

$$\begin{array}{r} 5 2 \\ \times 5 2 \\ \hline 2 5 \times 2 2 \times 2 \\ 5 \times 5 2 \times 5 \cancel{ 2 \times 2} \\ \hline 2 7 0 4 \\ \hline \downarrow \downarrow \downarrow \downarrow \\ P_4 P_2 P_1 P_0 \end{array}$$

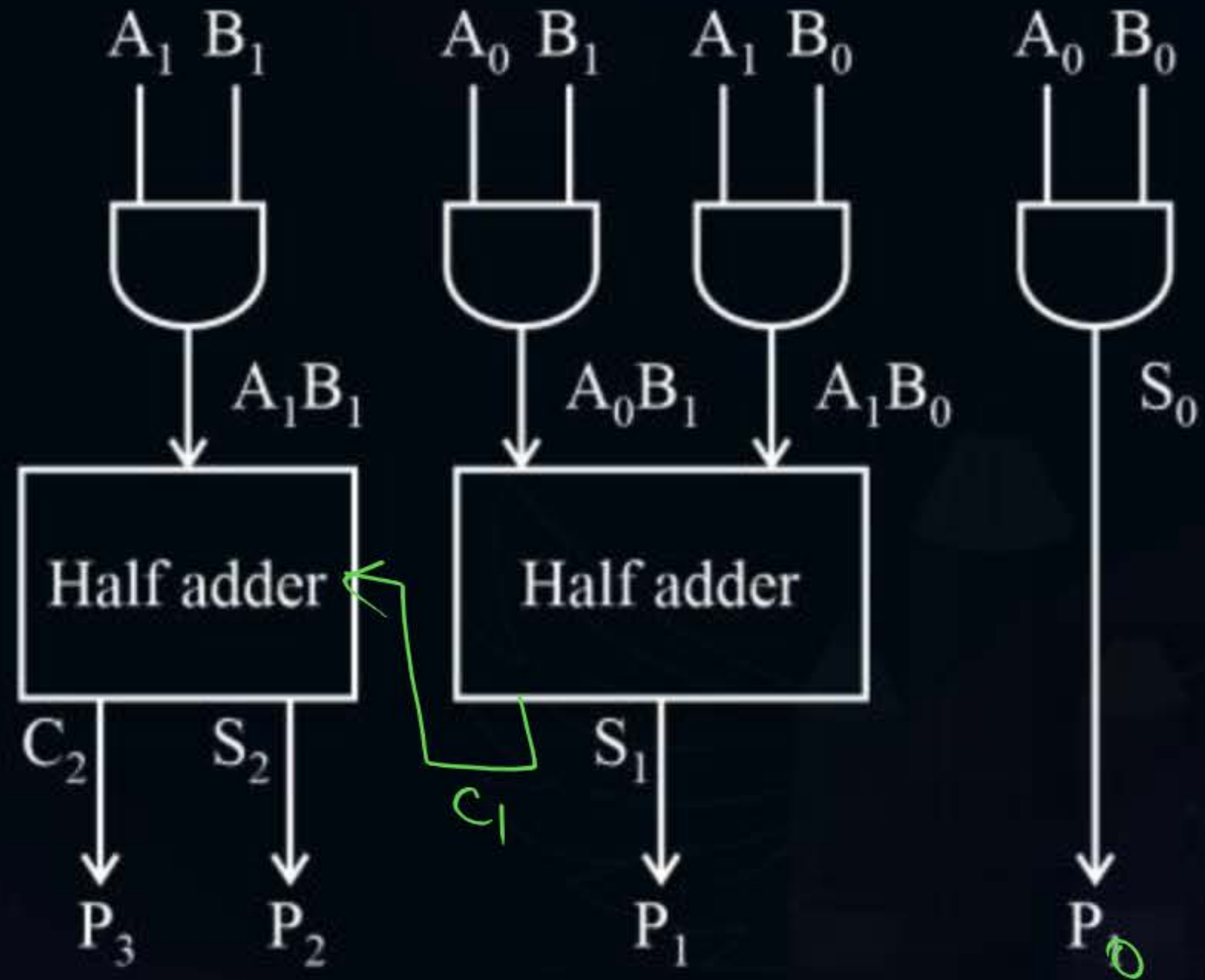


Topic : PARALLEL ADDER, LACA

EsKool
with

general

	A_1	A_0	
$\times$	B_1	B_0	
C_1	A_1B_0	A_0B_0	
A_1B_1	A_0B_1	$\times$	
C_2	S_2	S_1	S_0
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
P_3	P_2	P_1	P_0



#Q. Design a circuit in which 4 inputs A, B, C, D are there. if in decimal $AB = X$ and $CD = Y$. Then if $X.Y \leq 3$, Then o/p will be high other-wise Low.

~~$X.Y \leq 0$~~

$\leq$

Demorgan chadhaji

GO!

$F(A, B, C, D)$

$\overline{A+C} = \overline{A \cdot C}$

A) $A \cdot C$

B) $\overline{A \cdot C}$

C) $\overline{A+C}$

D) $A+C$

What is the Simplified Expression?

$AB \searrow \nearrow CD$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 0	1 1	1 3	1 2
$\bar{A}B$	1 4	1 5	1 7	1 6
AB	1 12	1 13		
$A\bar{B}$	1 8	1 9		

$\bar{A}$ (points to the first two columns)
 $\bar{C}$ (points to the first two rows)

$$\begin{aligned}
 &\Rightarrow F(A, B, C, D) \\
 &= \bar{A} + \bar{C} \\
 &= \overline{A \cdot C}
 \end{aligned}$$



Topic : PARALLEL ADDER, LACA

HA

$$S = A \oplus B$$

$$\text{Carry} = AB$$

5

NAND/
NOR

FA

$$S = A \oplus B \oplus C$$

$$\text{Carry} = \Sigma m(3, 5, 6, 7)$$

$$= (A \oplus B)C + AB$$

$$= AB + AC + BC$$

9

$$1. \text{FA} = 2 \text{MF} + 1 \text{ OR}$$

$$1 \text{ FS} = 2 \text{ FS} + 1 \text{ OR}$$

HS

$$\text{Diff} = A \oplus B$$

$$\text{Carry} = \bar{A}B$$

5

FS

$$\text{Diff} = A \oplus B \oplus C$$

$$\text{Carry} = \Sigma m(1, 2, 3, 7)$$

$$= \bar{A}B + (\overline{A \oplus B})C$$

$$= \bar{A}B + \bar{A}C + BC$$

9



Topic : PARALLEL ADDER, LACA

Serial Adder:

nT :

Parallel Adder:

$$T = (n - 1) T_{\text{Carry}} + \overset{\text{max}}{\cancel{T_{\text{max}}}} \{T_{\text{Sum}}, T_{\text{Carry}}\}$$

$$T = (n - 1) \{T_{\text{AND}} + T_{\text{OR}}\} + \text{MAX} \{T_{\text{Sum}}, T_{\text{Carry}}\}$$



Topic : PARALLEL ADDER, LACA

LACA:

$$\text{Total Delay} = \underline{\underline{4T}}$$

$$\text{Carry Block} = \underline{\underline{2T}}$$

Carry Block:

n-bit LACA

$$\text{AND} = \frac{n(n+1)}{2}$$

~~Carry Block = n~~

$$\text{OR gates} = n$$



2 mins Summary



Combination ckt

- Ripple Carry Adder (Parallel Adder)
- LACA
- Questions
- Multiplier



THANK - YOU